

Spotting the weird ones

Rob J Hyndman

6 August 2026

robjhyndman.com/ihaka2026



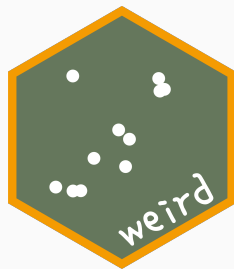
Thanks Ross!

- Ross Ihaka is my academic great uncle
- 1987: I first used S
- 1988: I switched to S-PLUS
- 1994: I heard Ross & Robert give a talk on R at ASC
- 2000: I switched to R
- 2006: My first CRAN package
- 2026: Coauthor of 63 R packages
- 2023–2026: Executive Editor, *R Journal*



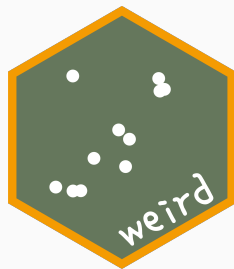
weird R package

- Functions and data sets required for *That's Weird: Anomaly Detection Using R*
[OTexts.com/weird](https://otexts.com/weird)
- All packages needed to run the examples are also loaded



weird R package

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```
library(weird)
```

```
— Attaching core weird packages — weird 3.0.0 —
✓ distributional 0.8.1
✓ ggplot2        4.0.3
✓ dplyr          1.2.1
— Conflicts — weird_conflicts —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
```

Outline

- 1 What is an anomaly?
- 2 Wine quality and prices
- 3 Approximate extreme surprisal probabilities
- 4 Cricket not-outs
- 5 Old Faithful eruptions
- 6 French mortality rates

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What do we mean by “weird”?

Monographs on Statistics and Applied Probability

D.M. Hawkins

Identification
of Outliers

an observation which deviates so much from other observations as to arouse suspicion it was generated by a different mechanism. (Hawkins, 1980)

What do we mean by “weird”?

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D.M. Hawkins

Identification
of Outliers

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V. BARNETT
T. LEWIS

OUTLIERS IN
STATISTICAL DATA

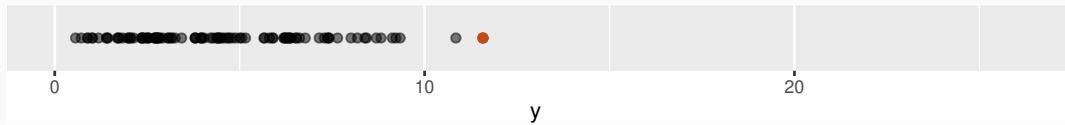
3RD EDITION

WILEY SERIES IN PROBABILITY
AND MATHEMATICAL STATISTICS

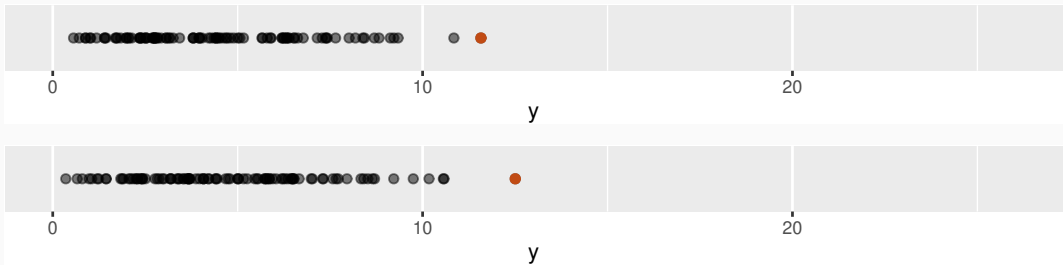


an observation (or a subset of observations) which appears to be inconsistent with the remainder of that set of data. (Barnett & Lewis, 3rd ed., 1994)

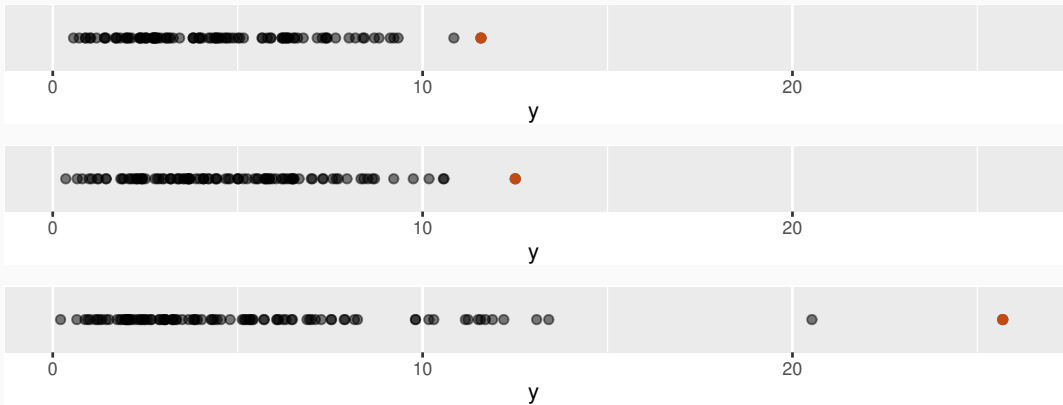
Is this an anomaly?



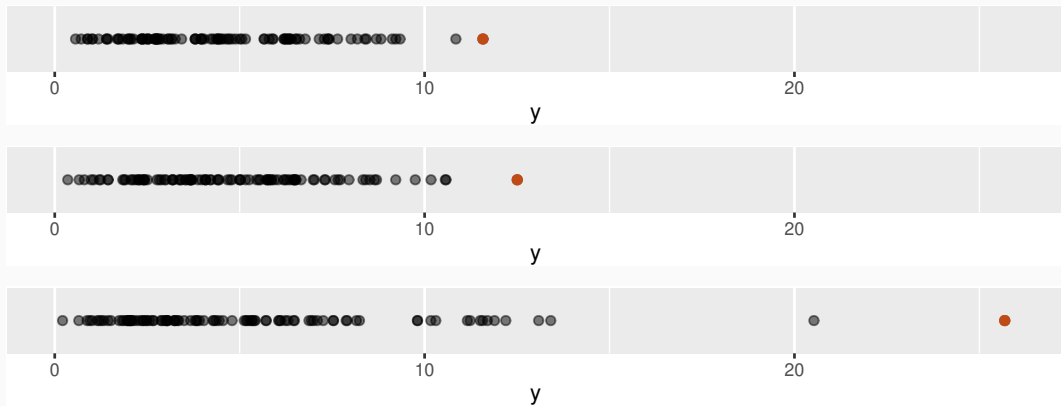
Is this an anomaly?



Is this an anomaly?

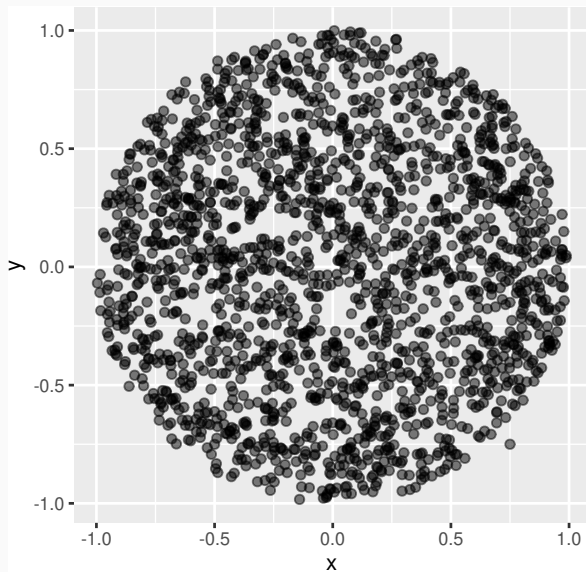


Is this an anomaly?

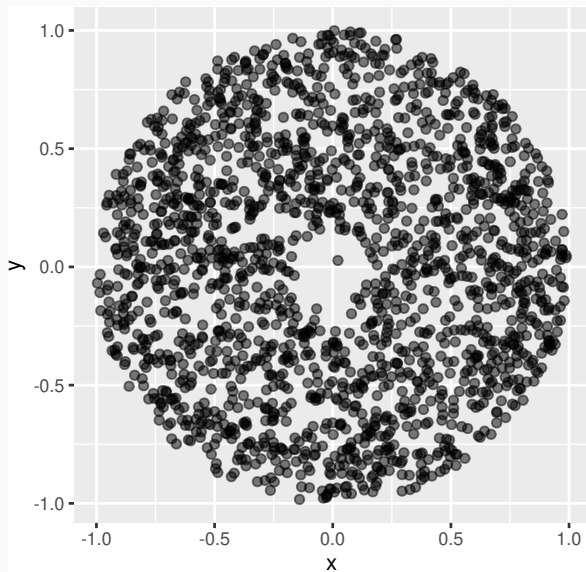


All points were randomly generated from the **same** χ^2_5 distribution.

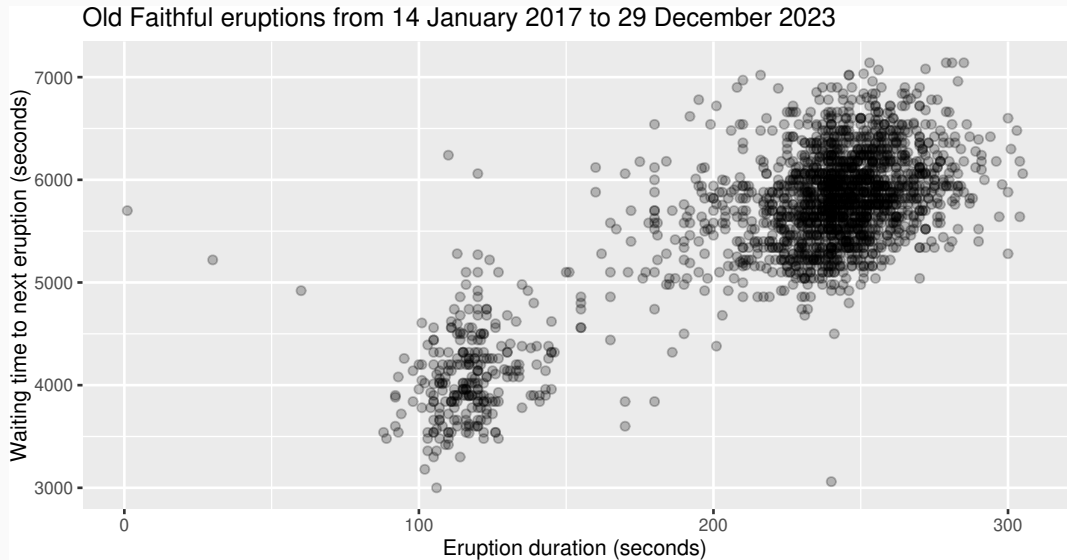
Spot the anomaly



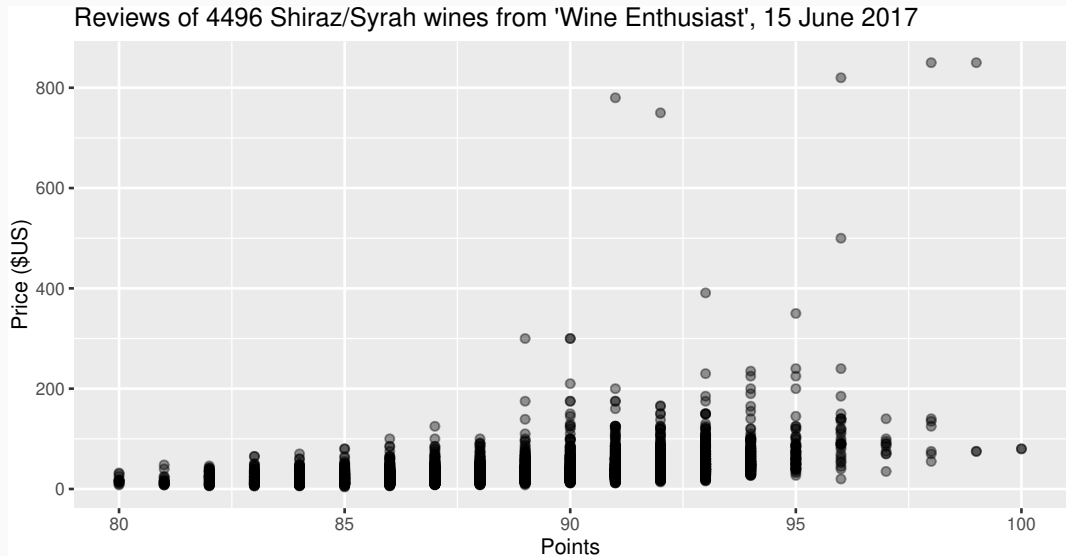
Spot the anomaly



Are there any anomalies here?



Are there any anomalies here?



Surprisal probabilities

Surprisal probability of observation y_i under model f

The probability of an observation that is at least as unlikely:

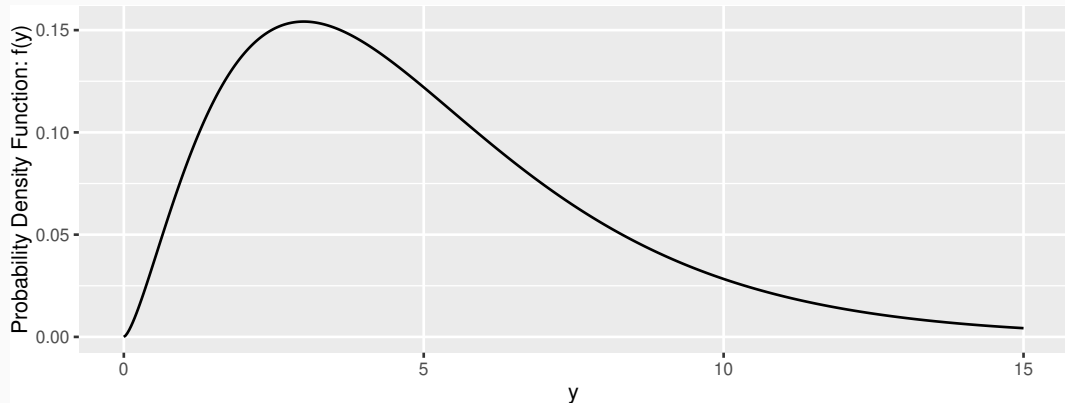
$$p_i = \mathbb{P}(f(Y) \leq f(y_i)) \quad \text{where } Y \text{ has density } f$$

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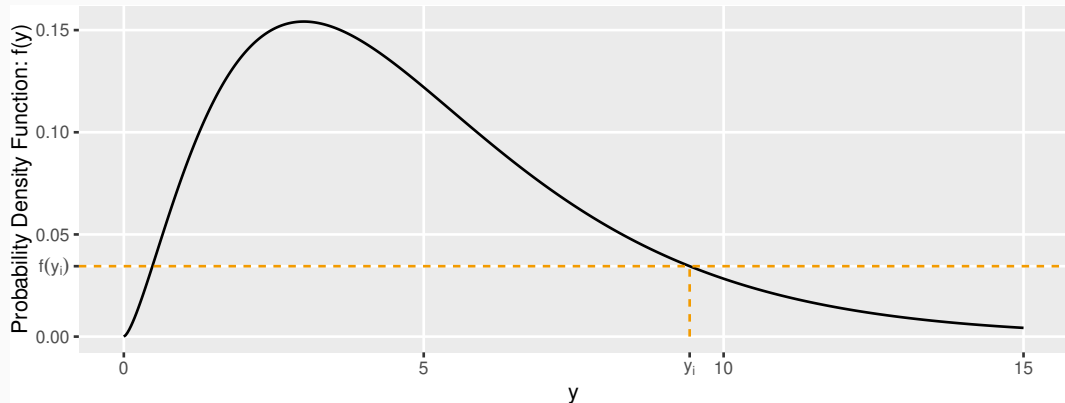


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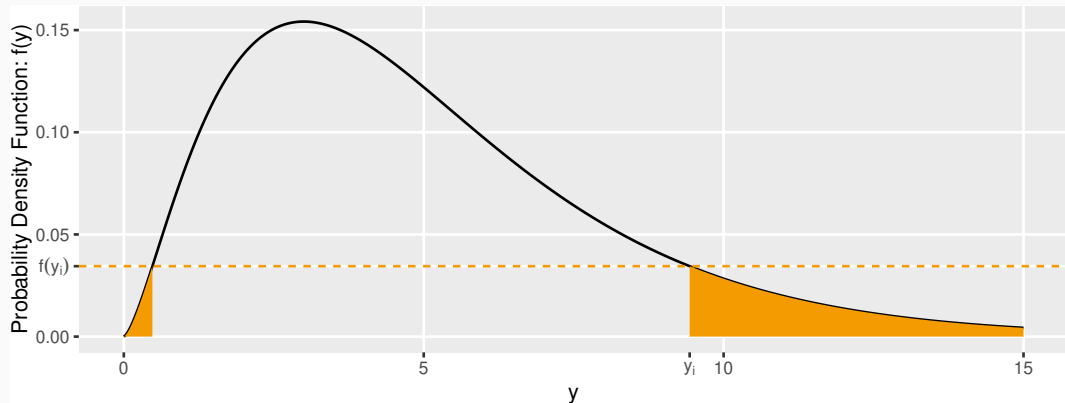


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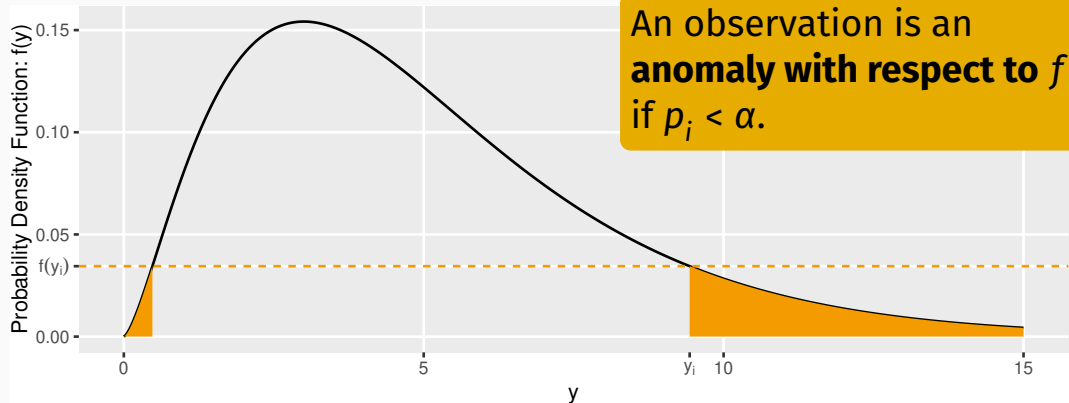


Surprisal probabilities

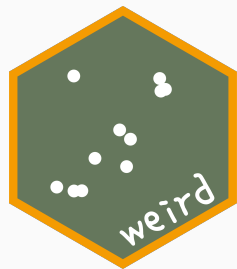
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weird R package



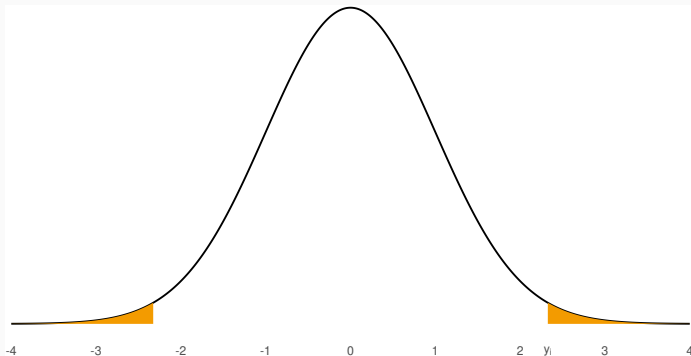
- `surprisals_prob( )` is an S3 generic that computes surprisal probabilities for any data object using a user-specified density function, or a kernel density estimate. It also works with linear models, GLMs, and GAMs.
- Distributions can be specified as distributional objects from the distributional package.

weird R package

```
N01 <- dist_normal(mean = 0, sd = 1)
tibble(
  y = generate(N01, times = 1000)[[1]],
  p = surprisals_prob(y, distribution = N01)
) |>
  arrange(p)
```

A tibble: 1,000 x 2

	y	p
	<dbl>	<dbl>
1	3.42	0.000634
2	3.14	0.00167
3	-3.05	0.00230
4	-2.96	0.00304
5	-2.89	0.00390
6	2.81	0.00488
7	2.64	0.00832
8	-2.60	0.00923
9	2.56	0.0104



Outline

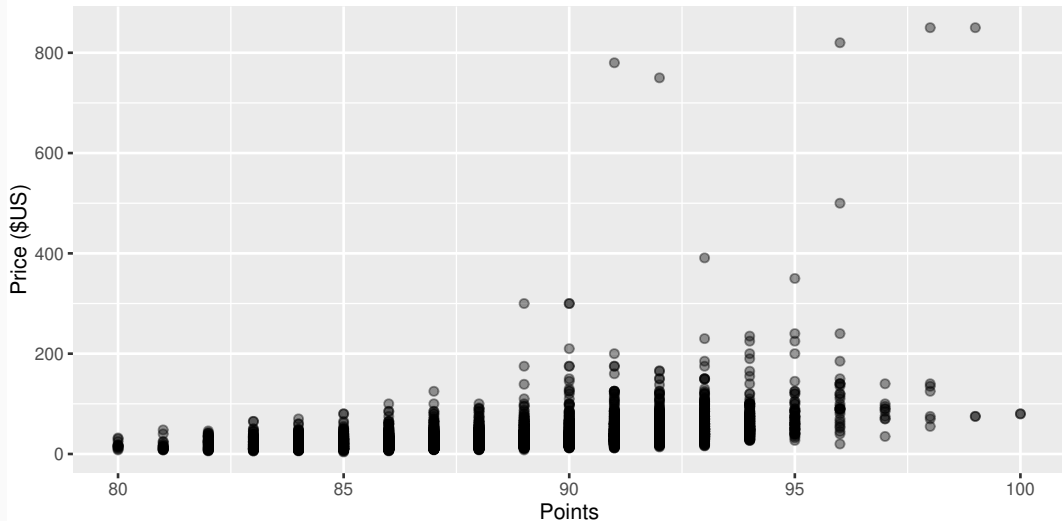
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Wine quality and prices

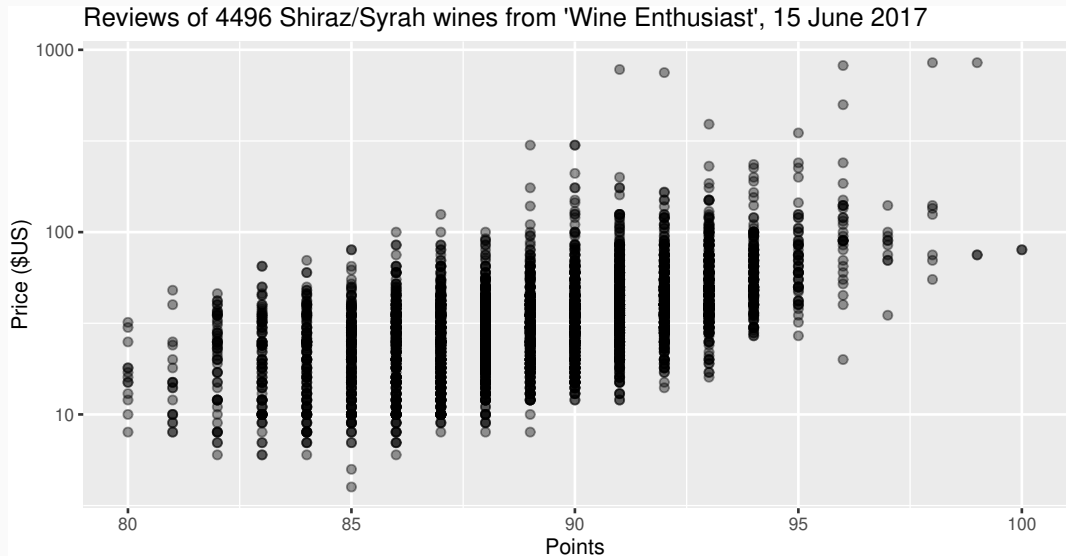


Wine quality and prices

Reviews of 4496 Shiraz/Syrah wines from 'Wine Enthusiast', 15 June 2017



Wine quality and prices

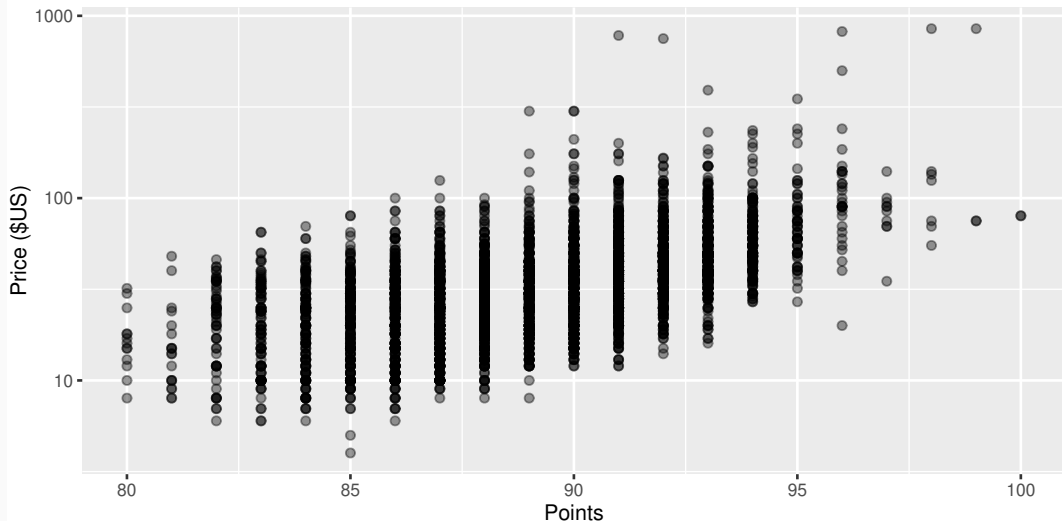


Wine quality and prices

Statistical model

$$\log(\text{Price}) \sim N(a + b\text{Points}, \sigma^2)$$

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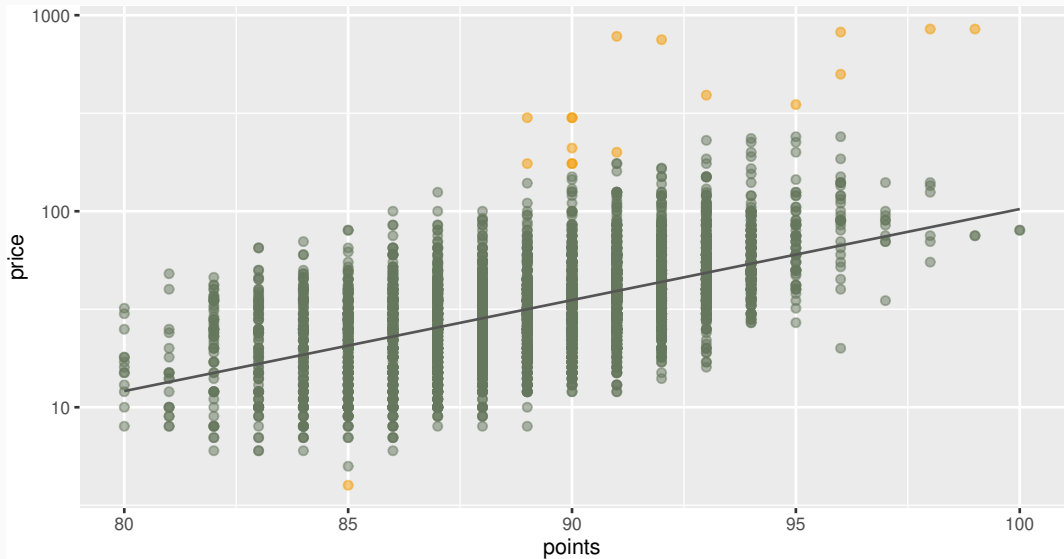
```
wine_lm <- lm(log(price) ~ points, data = wine_reviews)
wine_reviews <- wine_reviews |>
  mutate(p = surprisals_prob(wine_lm, loo = TRUE))
wine_reviews |> arrange(p)
```

A tibble: 4,496 x 7

	country	state	winery	year	points	price	p
	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Australia	South Australia	Henschke	2009	91	780	5.77e-10
2	US	California	Law	2013	92	750	3.82e- 9
3	Australia	South Australia	Henschke	2010	96	820	2.12e- 7
4	Australia	South Australia	Penfolds	2008	98	850	1.43e- 6
5	Italy	Tuscany	Tua Rita	2011	89	300	3.30e- 6
6	Australia	South Australia	Penfolds	2010	99	850	4.24e- 6
7	Italy	Tuscany	Tua Rita	2013	90	300	9.46e- 6
8	Italy	Tuscany	Tua Rita	2012	90	300	9.46e- 6
9	France	Rhône Valley	Domaine Jean-Michel Ge~	2013	93	391	1.60e- 5
10	Australia	South Australia	Penfolds	2004	96	500	3.18e- 5

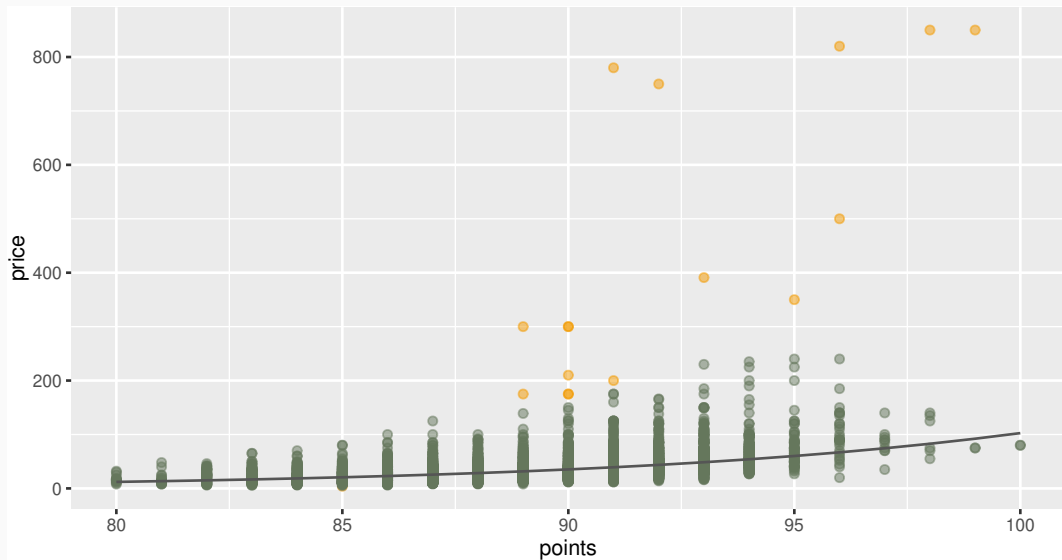
Wine quality and prices

$p_i < 0.001$



Wine quality and prices

$p_i < 0.001$



Weirdly low priced wines

```
wine_reviews |>  
  filter(log(price) < fitted(wine_lm)) |>  
  arrange(p, points, price)
```

```
# A tibble: 2,307 x 7
```

	country	state	winery	year	points	price	p
	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Spain	Central Spain	Felix Solis	2013	85	4	7.00e-4
2	Australia	Australia Other	Banrock Station	2006	85	5	3.42e-3
3	France	Languedoc-Roussillon	Domaine Borie de M~	1999	89	8	4.52e-3
4	US	Washington	Two Vines	2012	86	6	5.58e-3
5	US	Washington	StoneCap	2010	88	8	8.84e-3
6	US	Washington	Rulo	2007	96	20	1.27e-2
7	Australia	Australia Other	Yellow Tail	2001	86	7	1.42e-2
8	France	Languedoc-Roussillon	Calvet	1998	86	7	1.42e-2
9	Chile	Central Valley	Morandé	2002	86	7	1.42e-2
10	India	Nashik	Sula	2012	91	12	1.46e-2

```
# i 2,297 more rows
```


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- **Surprisal**: $s_i = -\log f(y_i)$. Anomalies are extreme values in surprisal space.

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- We can use **extreme value theory** when f is wrong: the upper tail of the distribution of s_i is approximately Generalized Pareto Distribution, even if f is incorrect

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- **Surprisal**: $s_i = -\log f(y_i)$. Anomalies are extreme values in surprisal space.
- We can use **extreme value theory** when f is wrong: the upper tail of the distribution of s_i is approximately Generalized Pareto Distribution, even if f is incorrect
- So $p_i = \mathbb{P}(-\log f(Y) \geq s_i) \approx 1 - F_{GPD}(s_i)$

Wine quality and prices

```
wine_reviews <- wine_reviews |>
  mutate(p = surprisals_prob(wine_lm, loo = TRUE, approximation = "gpd"))
wine_reviews |> arrange(p)
```

A tibble: 4,496 x 7

	country	state	winery	year	points	price	p
	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Australia	South Australia	Henschke	2009	91	780	1.03e-4
2	US	California	Law	2013	92	750	1.39e-4
3	Australia	South Australia	Henschke	2010	96	820	2.94e-4
4	Australia	South Australia	Penfolds	2008	98	850	4.52e-4
5	Italy	Tuscany	Tua Rita	2011	89	300	5.56e-4
6	Australia	South Australia	Penfolds	2010	99	850	5.93e-4
7	Italy	Tuscany	Tua Rita	2013	90	300	7.37e-4
8	Italy	Tuscany	Tua Rita	2012	90	300	7.37e-4
9	France	Rhône Valley	Domaine Jean-Michel Ger~	2013	93	391	8.56e-4
10	Australia	South Australia	Penfolds	2004	96	500	1.05e-3

i 4,486 more rows

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Cricket not-outs



Cricket not-outs

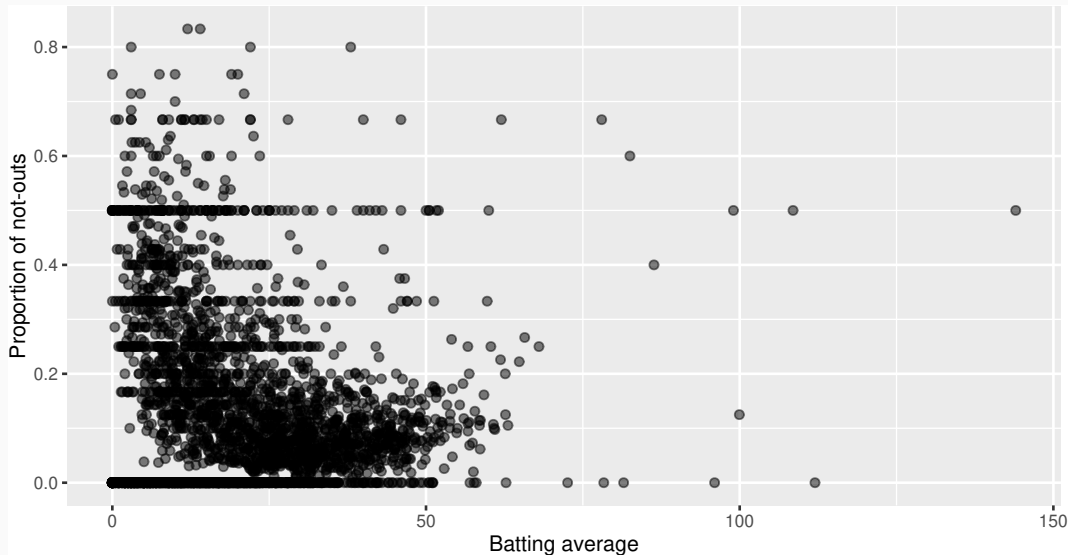
```
df_no <- cricket_batting |>
  filter(Innings > 0, Innings > NotOuts, Gender == "Men") |>
  mutate(prop_no = NotOuts / Innings)
df_no
```

A tibble: 3,144 x 16

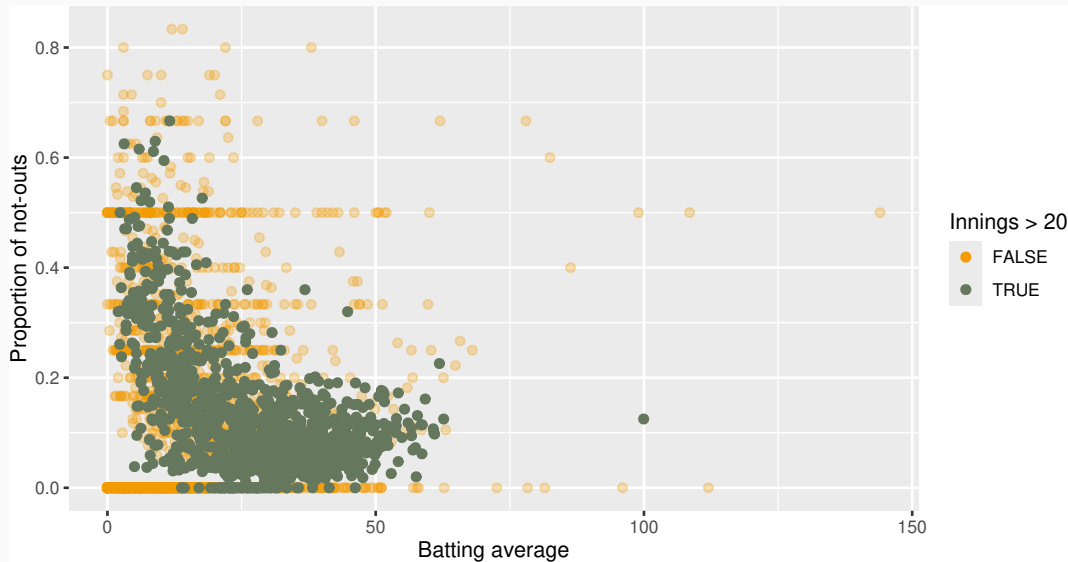
	Player	Country	Start	End	Matches	Innings	NotOuts	Runs	HighScore
	<chr>	<chr>	<int>	<int>	<int>	<int>	<int>	<int>	<int>
1	BB Cooper	Australia	1877	1877	1	2	0	18	15
2	C Bannerman	Australia	1877	1879	3	6	2	239	165
3	DW Gregory	Australia	1877	1879	3	5	2	60	43
4	EJ Gregory	Australia	1877	1877	1	2	0	11	11
5	FR Spofforth	Australia	1877	1887	18	29	6	217	50
6	JM Blackham	Australia	1877	1894	35	62	11	800	74
7	JR Hodges	Australia	1877	1877	2	4	1	10	8
8	NFD Thomson	Australia	1877	1877	2	4	0	67	41
9	TJD Kelly	Australia	1877	1879	2	3	0	64	35
10	TK Kendall	Australia	1877	1877	2	4	1	39	17

i 3,134 more rows

Cricket not-outs



Cricket not-outs

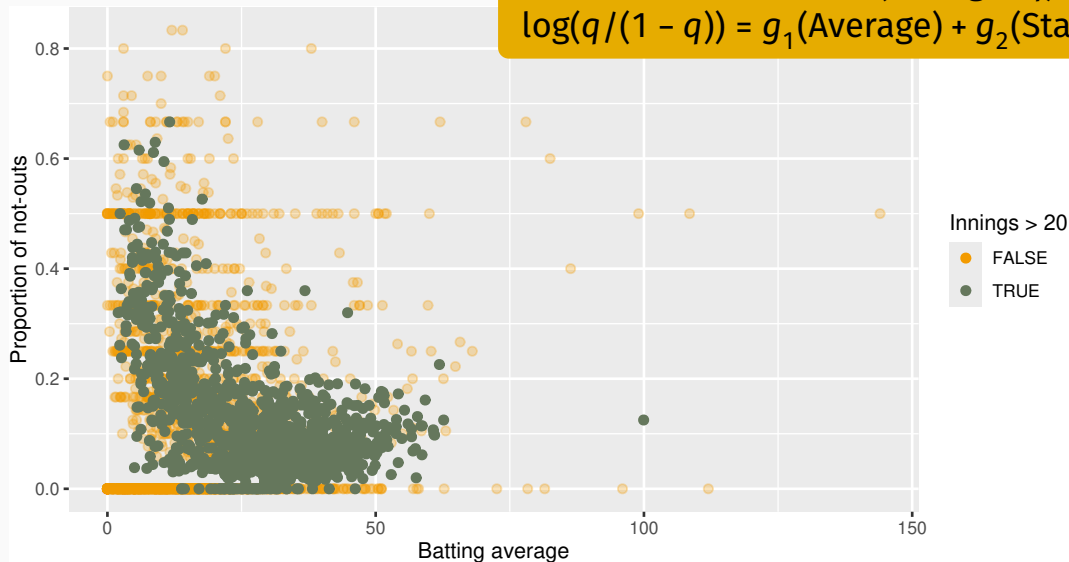


Cricket not-outs

Statistical model

NotOuts $\sim$ Binomial(Innings, q)

$\log(q/(1 - q)) = g_1(\text{Average}) + g_2(\text{Start})$



Cricket not-outs

```
fit_notouts <- mgcv::gam(  
  prop_no ~ s(Average) + s(Start),  
  data = df_no, weights = Innings,  
  family = binomial(link = logit),  
)
```

Statistical model

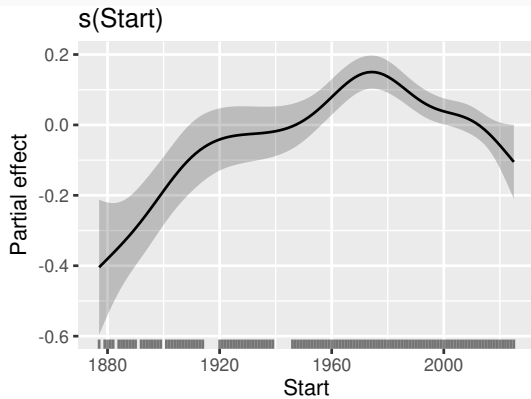
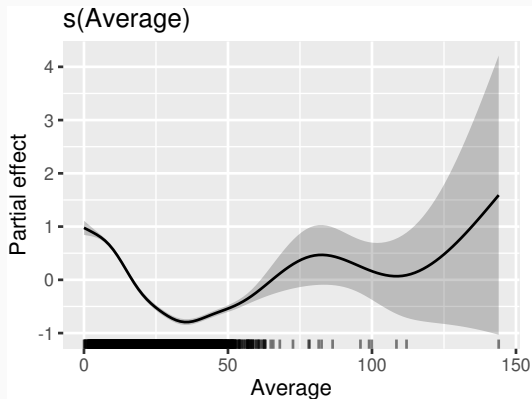
NotOuts ~ Binomial(Innings, q)
 $\log(q/(1 - q)) = g_1(\text{Average}) + g_2(\text{Start})$

Cricket not-outs

Statistical model

$$\text{NotOuts} \sim \text{Binomial}(\text{Innings}, q)$$
$$\log(q/(1 - q)) = g_1(\text{Average}) + g_2(\text{Start})$$

```
fit_notouts <- mgcv::gam(  
  prop_no ~ s(Average) + s(Start),  
  data = df_no, weights = Innings,  
  family = binomial(link = logit),  
)
```



Cricket not-outs

```
augment(fit_notouts, data = df_no) |>  
  mutate(p = surprisals_prob(fit_notouts, approximation = "gpd")) |>  
  select(Player:End, Innings, NotOuts, Average, p) |>  
  arrange(p)
```

A tibble: 3,144 x 8

	Player	Country	Start	End	Innings	NotOuts	Average	p
	<chr>	<chr>	<int>	<int>	<int>	<int>	<dbl>	<dbl>
1	TA Boult	New Zealand	2011	2022	94	46	15.8	0.000180
2	SM Pollock	South Africa	1995	2008	156	39	32.3	0.000452
3	JM Anderson	England	2003	2024	265	114	8.96	0.000621
4	JR Hazlewood	Australia	2014	2025	96	47	11.5	0.000812
5	A D'Souza	Pakistan	1959	1962	10	8	38	0.00148
6	Tauseef Ahmed	Pakistan	1980	1993	38	20	17.7	0.00175
7	FJ Cameron	New Zealand	1961	1965	30	20	11.6	0.00231
8	EJ Chatfield	New Zealand	1975	1989	54	33	8.57	0.00236
9	RR Jumadeen	West Indies	1972	1979	14	10	21	0.00257
10	W Rhodes	England	1899	1930	98	21	30.2	0.00286

i 3,134 more rows

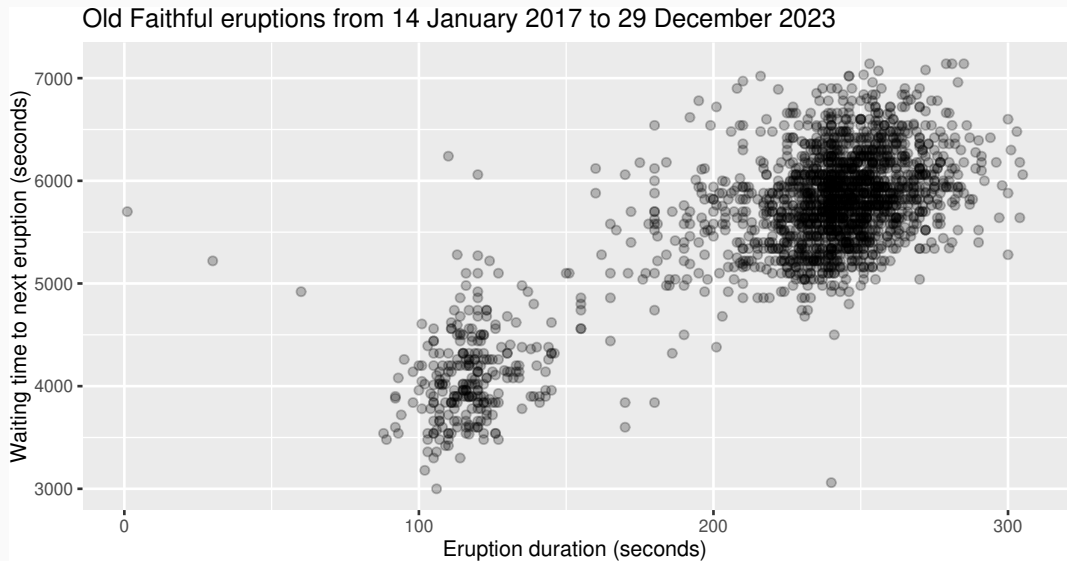
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Old Faithful eruptions



Old Faithful eruptions



Old Faithful eruptions

```
oldfaithful
```

```
# A tibble: 2,097 x 4
```

	time	recorded_duration	duration	waiting
	<dtm>	<chr>	<dbl>	<dbl>
1	2017-01-14 00:06:00	3m 16s	196	5940
2	2017-01-26 14:27:00	~4m	240	5820
3	2017-01-27 23:57:00	2m 1s	121	3900
4	2017-01-30 15:09:00	~4m	240	5280
5	2017-01-31 13:27:00	~3.5m	210	5580
6	2017-01-31 15:00:00	~4m	240	5760
7	2017-02-03 23:13:00	3m 25s	205	5160
8	2017-02-04 22:14:00	3m 34s	214	5400
9	2017-02-05 17:19:00	4m 0s	240	6060
10	2017-02-05 19:00:00	4m 2s	242	6060

```
# i 2,087 more rows
```

Downloaded from geysertimes.org

Old Faithful eruptions

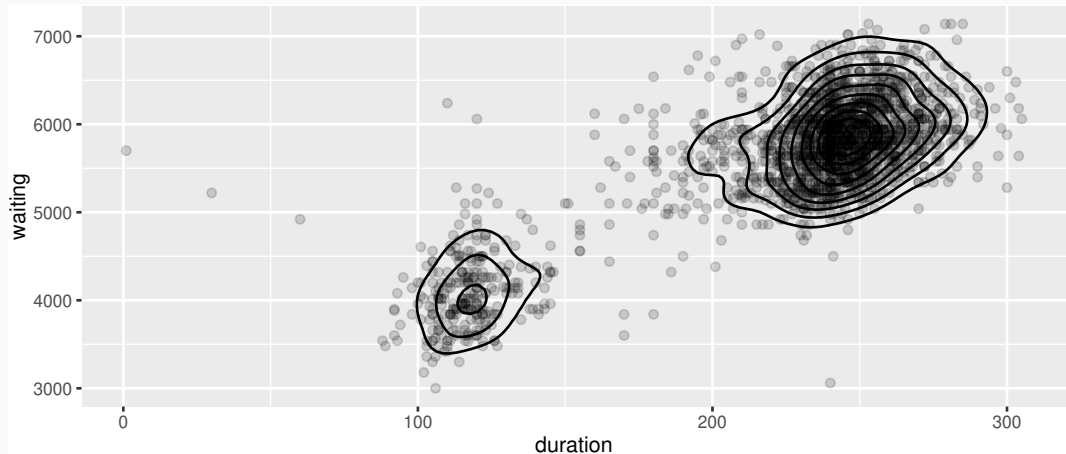
```
oldfaithful$p <- surprisals_prob(  
  oldfaithful[, c("duration", "waiting")],  
  loo = TRUE,  
  approximation = "gpd"  
)  
oldfaithful |> arrange(p)
```

A tibble: 2,097 x 5

	time	recorded_duration	duration	waiting	p
	<dtm>	<chr>	<dbl>	<dbl>	<dbl>
1	2018-04-25 19:08:00	1s	1	5700	0
2	2020-06-01 21:04:00	2 minutes	120	6060	0
3	2020-09-04 01:38:00	>1m 50s	110	6240	0
4	2022-12-03 16:20:00	~4m	240	3060	0
5	2022-12-07 17:19:00	~4 30s	30	5220	0
6	2023-07-04 12:03:00	~1 minute 55ish seconds	60	4920	0
7	2023-07-23 21:57:00	~3 minutes 40ish seconds	180	6540	0.000817
8	2022-09-08 05:13:00	5 min	300	5280	0.000853
9	2021-01-22 18:35:00	2m50s	170	3600	0.000878

Old Faithful eruptions

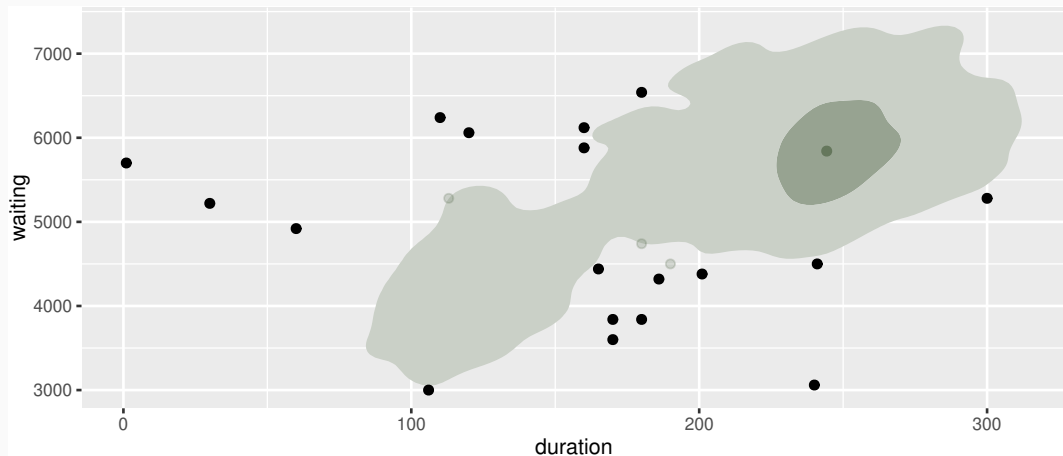
```
oldfaithful[, c("duration", "waiting")] |>  
  dist_kde() |>  
  gg_density(show_points = TRUE)
```



Old Faithful eruptions

$\alpha = 0.005$

```
oldfaithful |>  
  gg_hdrboxplot(duration, waiting)
```



Outline

- 1 What is an anomaly?
- 2 Wine quality and prices
- 3 Approximate extreme surprisal probabilities
- 4 Cricket not-outs
- 5 Old Faithful eruptions
- 6 French mortality rates

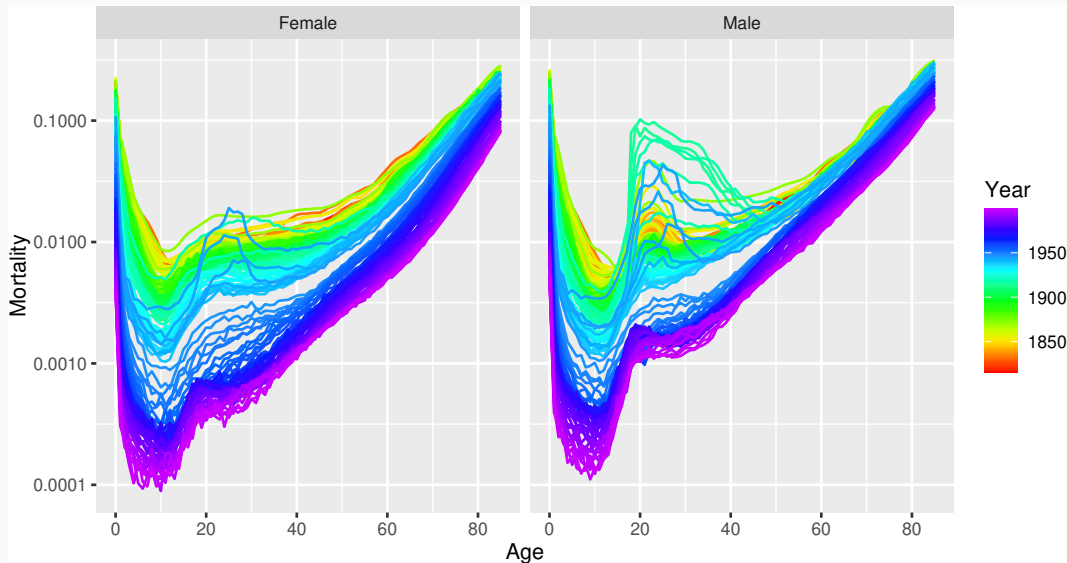
French mortality rates



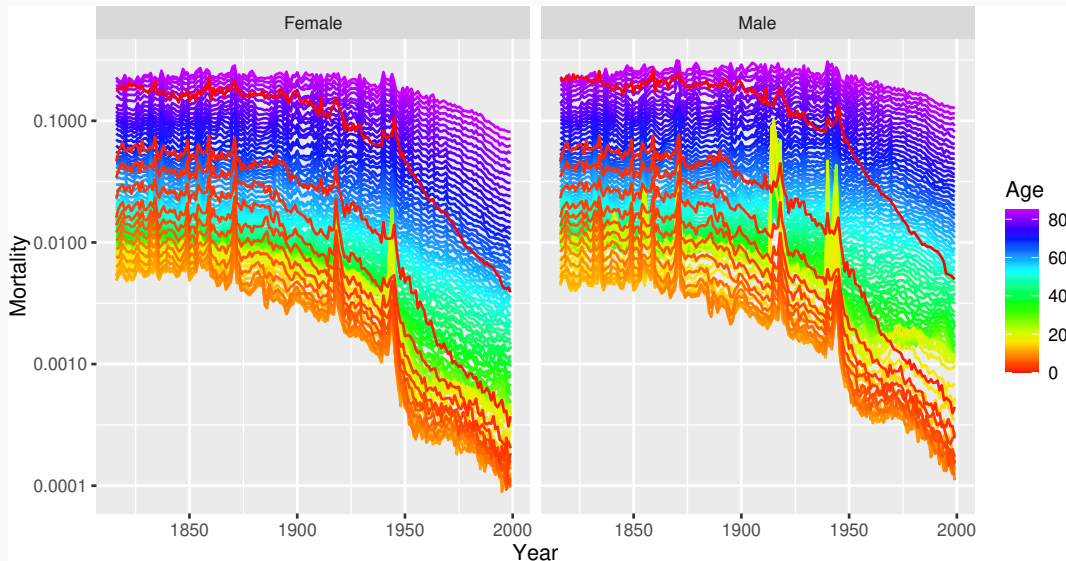
French mortality rates

```
# A tibble: 31,648 x 4
  Year   Age Sex   Mortality
  <int> <int> <chr>    <dbl>
1  1816     0 Female  0.187
2  1816     1 Female  0.0467
3  1816     2 Female  0.0339
4  1816     3 Female  0.0229
5  1816     4 Female  0.0160
6  1816     5 Female  0.0138
7  1816     6 Female  0.0121
8  1816     7 Female  0.0104
9  1816     8 Female  0.00891
10 1816     9 Female  0.00760
# i 31,638 more rows
```

French mortality rates



French mortality rates



French mortality rates

Hampel identifier

$$m_t = \text{median}(y_{t-h}, \dots, y_{t+h})$$

$$a_t = \text{median}(|y_{t-h} - m_t|, \dots, |y_{t+h} - m_t|)$$

$$e_t = |y_t - m_t| / a_t$$

- Anomaly if $e_t > k/0.6745$

French mortality rates

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- Implicit model: $y_t \sim N(m_t, \sigma_t^2)$

French mortality rates

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- Anomaly if $e_t > k/0.6745$
- Implicit model: $y_t \sim N(m_t, \sigma_t^2)$
- Anomaly if y_t is more than $k\sigma_t$ from m_t
- Equivalent to surprisal probability threshold of $\alpha = 0.0027$ for $k = 3$
- e_t values are not independent

French mortality rates

```
fr_anomalies <- fr_mortality |>
  group_by(Age, Sex) |>
  mutate(
    hampel = hampel_anomalies(
      Mortality,
      bandwidth = 7,
      alpha = 0.01,
      approximation = "gpd"
    )
  ) |>
  ungroup() |>
  filter(hampel)
```

French mortality rates

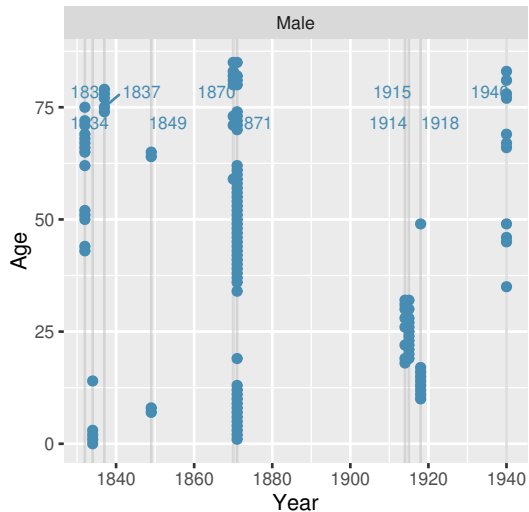
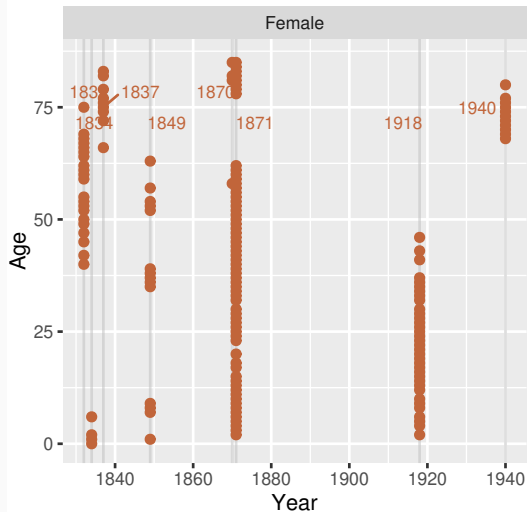
```
fr_anomalies <- fr_mortality |>
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  mutate(
    hampel = hampel_anomalies(
      Mortality,
      bandwidth = 7,
      alpha = 0.01,
      approximation = "gpd"
    )
  ) |>
  ungroup() |>
  filter(hampel)
```

```
fr_anomalies |>
  count(Year, Sex) |>
  filter(n >= 3)
```

	Year	Sex	n
1	1832	Female	21
2	1832	Male	14
3	1834	Female	4
4	1834	Male	5
5	1837	Female	9
6	1837	Male	5
7	1849	Female	14
8	1849	Male	4
9	1870	Female	4
10	1870	Male	8
11	1871	Female	64
12	1871	Male	51
13	1914	Male	8
14	1915	Male	12
15	1918	Female	35
16	1918	Male	9
17	1940	Female	11
18	1940	Male	11

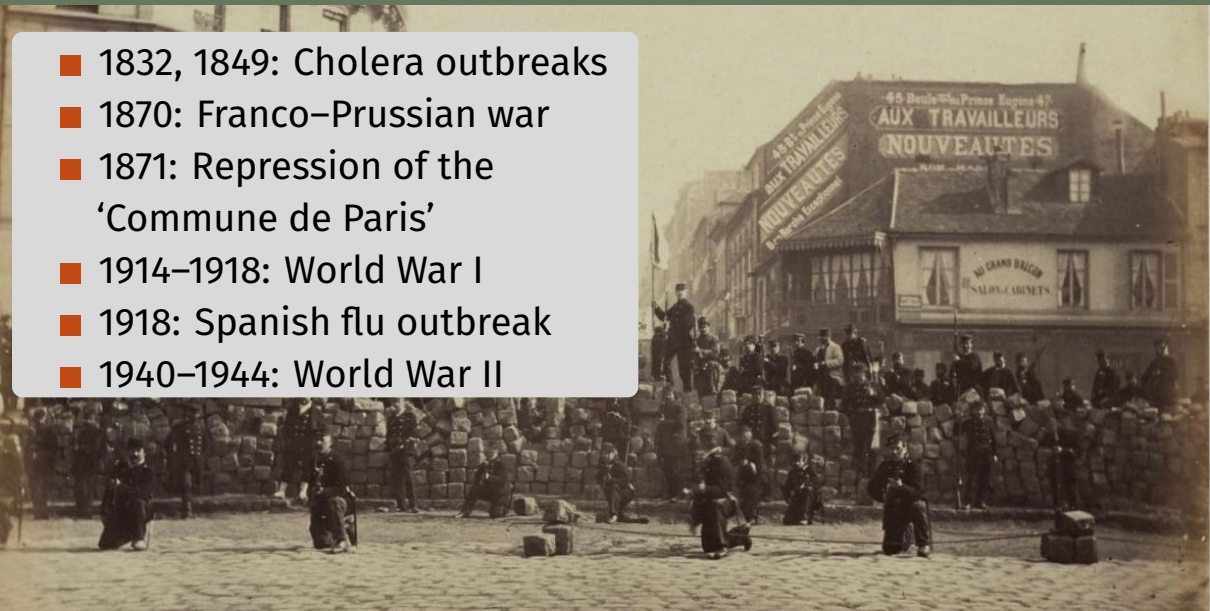
French mortality rates

French mortality anomalies



French mortality rates

- 1832, 1849: Cholera outbreaks
- 1870: Franco-Prussian war
- 1871: Repression of the 'Commune de Paris'
- 1914–1918: World War I
- 1918: Spanish flu outbreak
- 1940–1944: World War II



French mortality rates

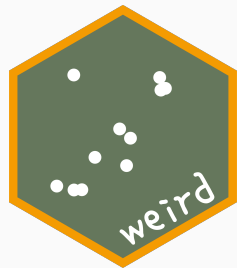
- 1832, 1849: Cholera outbreaks
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But what happened to:

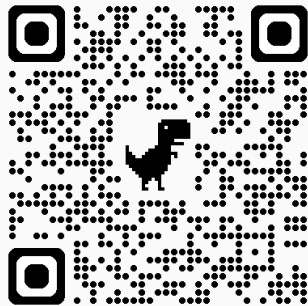
- infant mortality in 1834?
- elder mortality in 1837?



Find out more



- R package **weird** available on CRAN
- *That's weird* book in preparation at OTexts.com/weird
- Slides, links and related papers: robjhyndman.com/ihaka2026



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